

Yashwanth Kasanneni

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EDUCATION

NYU Tandon School of Engineering

Master's in Computer Science

Coursework: Machine Learning, Analysis of Algorithms, Distributed Systems, Principles of Database Systems

Expected May 2027

New York, United States

WORK EXPERIENCE

Software Developer Intern

Sept 2023 - Dec 2023

Enabled Analytics

On-Site

- Developed a custom Salesforce-based order management system integrated with WhatsApp, allowing supermarket customers to browse items, place orders, and receive automated responses through a chat-based workflow.
- Built Apex, SOQL/SOSL, and Salesforce Flow automation to process WhatsApp orders in real time, validate item availability, update inventory, prevent duplicate orders, and sync order data across Salesforce records.
- Created Salesforce Lightning interfaces for supermarket staff to monitor WhatsApp orders, update fulfillment status, manage inventory, and access customer/order history from a centralized dashboard, owning the feature end-to-end from design through testing and production deployment while debugging workflow-execution issues across order placement and inventory sync.

PROJECTS

Image-to-SVG Generation via Fine-Tuned and Distilled Vision-Language Model | *Python, PyTorch, Hugging Face Transformers, Qwen2-VL*

- Fine-tuned Qwen2-VL-7B-Instruct to generate SVG markup directly from an image input for a deep learning course competition, framing the task as sequence generation over valid SVG path and shape commands and evaluating by rendering the generated markup and measuring structural similarity (SSIM) against the target image on a held-out validation set.
- Compressed the fine-tuned 7B model under the competition's compute and latency budget using knowledge distillation into a 2B-parameter Qwen2-VL student, matching the teacher's output-token distribution over SVG sequences — the same distillation principle behind DistilBERT, which I'd used as an off-the-shelf checkpoint in my published NLP research — and the distilled student still reached 93% average SSIM against target images, retaining nearly all of the teacher's rendering fidelity at a fraction of the inference cost.

SpecDraft - Draft-Model Training for Speculative Decoding | *Python, PyTorch, Hugging Face Transformers*

- Trained a small draft model to approximate Qwen2-VL-7B-Instruct's token-level output distribution on the same SVG-generation task, then implemented speculative decoding — the draft model proposes several tokens at once, and the full model verifies them in parallel via rejection sampling — to speed up inference without changing the target model's actual output, unlike distillation, which trades accuracy for a smaller standalone model.
- Tuned the number of speculatively-proposed tokens per step to balance acceptance rate against wasted compute on rejected proposals, measured real wall-clock speedup on the same hardware and task used to benchmark the distilled student, and compared where each technique wins: speculative decoding when exact output fidelity matters, distillation when a smaller standalone model is needed.

CourtIQ - Real-Time Feature Pipeline (Lambda Architecture) | *Python, Kafka, PySpark Structured Streaming, Airflow, Apache Iceberg, PostgreSQL, FastAPI, React*

- Built a real-time data pipeline using a Lambda architecture — a Kafka event-streaming speed layer with PySpark Structured Streaming computing live features alongside an Airflow-orchestrated batch layer writing engineered features to Apache Iceberg tables — the same batch-plus-streaming pattern used to keep an ML feature store fresh without recomputing historical features from scratch.
- Served the merged real-time and batch feature views through a FastAPI backend and PostgreSQL, ran the full multi-service stack in Docker Compose, and conducted a 24-point architectural gap analysis of the pipeline's failure modes and scaling limits.

PolicyDiff - AI-Powered Payer Policy Change Detection for Denial Prevention | *Python, TypeScript, Next.js, FastAPI, Gemini 2.5 Flash, ClickHouse, Datadog LLM Observability*

- Co-built a three-microservice pipeline that translates a real business problem — catching harmful healthcare-policy changes before they cause claim denials — into a structured classification task, using Gemini 2.5 Flash with a constrained output schema to classify each change and a SHA-1 diff step to generate labeled training-relevant examples from real policy document versions.
- Evaluated classification quality against a hand-labeled validation set, flagged low-confidence predictions for mandatory human review instead of auto-publishing them, and instrumented every model call through Datadog LLM Observability to monitor prediction quality and latency once the model was serving real production traffic.

RESEARCH AND PUBLICATIONS

[Effective Analysis of Machine and Deep Learning Methods for Diagnosing Mental Health Using Social Media Conversations](#)

IEEE Transactions on Computational Social Systems

- Analyzed 100K+ tweets and developed NLP pipelines using BERT, RoBERTa, DistilBERT, XLNet, and classical/deep-learning models for behavioral risk-pattern detection.
- Evaluated model accuracy and sentiment features to build predictive systems that identified psychological-condition signals with 97% accuracy.

SKILLS

Languages and Frameworks: Java, Python, SQL, JavaScript/TypeScript, React, Next.js, FastAPI, data structures and algorithms

ML Training and Evaluation: PyTorch, Hugging Face Transformers, fine-tuning, knowledge distillation, speculative decoding, rejection sampling, model evaluation

Data Engineering: PySpark, PySpark Structured Streaming, Kafka (event streaming), Airflow, Apache Iceberg, PostgreSQL, ClickHouse

AI Applied to Production Systems: LLM integration (Gemini), structured output schemas, LLM observability (Datadog), NLP

Certifications: Machine Learning Specialization by Andrew Ng, Google Cloud Digital Leader